MR-1: Lorentzian Continuation of Spectral Causal Theory

Wick Rotation, Complex Zeros, and Spectral Structure
of the Modified Graviton Propagator

David Alfyorov

March 2026

Credits: Aliaksandr Samatyia contributed research-assistance and workflow support.

Abstract

We perform the Lorentzian continuation of the one-loop SCT spectral action established in NT-1/NT-1b/NT-2/NT-4a. The master function $\varphi(z)$ and its analytic continuation to the Lorentzian axis $z = -x_L$ are derived in closed form and cross-validated to ∼50 digits by three independent methods. We construct the Lorentzian propagator denominators Π_{TT} (spin-2) and Π_s (spin-0), locate their complex zeros via grid-based Newton search and validate zero counts using the argument principle. Within $|z| < 20$, we find exactly two zeros of Π_{TT}: a Type A Euclidean ghost at $z_0 \approx 2.4148$ with residue $R_2 \approx -0.4931$, and a Type B Lorentzian ghost at $z_L \approx 1.2807$ with residue $R_L \approx -0.5378$. For the spin-0 sector at $\xi = 0$, we find a conjugate pair (Type C) at $z \approx -2.076 \pm 3.184\,\mathrm{i}$; at conformal coupling $\xi = 1/6$ the scalar mode decouples identically. No Type D (anomalous) zeros are found. The Källén–Lehmann spectral representation is set up for subsequent unitarity analysis (MR-2). Sub-tasks (a) and (d) on Krein-space formulation and causal-set constraints are treated at the level of a literature review with explicit gap markers.

Contents

1 Introduction and Conventions **3**

2 Sub-task (a): Krein Space Formulation **3**
2.1 Motivation . . . 3
2.2 Krein Space Structure . . . 3
2.3 Spectral Action in Krein Space . . . 3
2.4 Implications for SCT . . . 4

3 Sub-task (b): Lorentzian D^2 Analysis **4**
3.1 Euclidean vs. Lorentzian Laplacian . . . 4
3.2 Heat Kernel on the Lorentzian Side . . . 5
3.3 Form Factor Continuation . . . 5

4 Sub-task (c): Wick Rotation of Form Factors **5**
4.1 The Master Function on the Lorentzian Axis . . . 5
4.2 Per-Spin Form Factors on the Lorentzian Axis . . . 6
4.3 Lorentzian Propagator Denominators . . . 7
4.4 Lorentzian Ghost Zero . . . 8
4.5 Complex Zero Catalogue . . . 9

4.5.1 Classification Scheme . . . 9
4.5.2 Results: $\Pi_{\mathrm{TT}}(z)$ . . . 9
4.5.3 Results: $\Pi_s(z, \xi = 0)$ . . . 9
4.5.4 Comparison with Stelle Gravity . . . 10
4.6 Källén–Lehmann Spectral Representation . . . 11

5 Sub-task (d): Causal Set Constraints 11
5.1 Motivation . . . 11
5.2 Causal Set Propagator . . . 12
5.3 Constraints on SCT . . . 12

6 Summary of Results 12
6.1 Key Formulas . . . 12
6.2 Verification Summary . . . 13
6.3 Gap Summary . . . 13

7 Outlook 13

1 Introduction and Conventions

This document presents the derivation of the Lorentzian continuation (MR-1) of Spectral Causal Theory. It builds on the following established results:

- **NT-1/NT-1b** (Phases 1–3): one-loop form factors $h_C^{(s)}(z)$, $h_R^{(s)}(z)$ for spins $s = 0, 1/2, 1$, and the total coefficients $\alpha_C = 13/120$, $\alpha_R(\xi) = 2(\xi - 1/6)^2$.
- **NT-2**: the entire-function property of $\varphi(z)$, $F_1(z)$, and $F_2(z, \xi)$.
- **NT-4a**: linearized field equations with propagator denominators $\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z)$ and $\Pi_s(z, \xi)$.
- **MR-2 d.5**: Euclidean residue analysis confirming the ghost pole at $z_0 \approx 2.4148$ with $R_2 \approx -0.4931$.

Convention 1.1 (Sign and signature conventions)**.** Throughout this document:

1. Euclidean signature $(+, +, +, +)$ for the spectral-action computation. Lorentzian signature $(-, +, +, +)$.
2. $\Box_E = -\partial_\mu \partial_\mu$ (positive-definite Euclidean Laplacian); $\Box_L = -\partial_t^2 + \partial_i \partial_i$ (Lorentzian d'Alembertian). Relation: $\Box_E \to -\Box_L$.
3. Dimensionless variable $z = \Box/\Lambda^2$ (Euclidean: $z > 0$ for spacelike); $z_L = k^2/\Lambda^2$ (Lorentzian: $z_L > 0$ for timelike).
4. **Wick rotation:** $z_E \to z_L = -z_E$, so evaluating a Euclidean form factor at Lorentzian momentum corresponds to $f(z_E) \to f(-z_L)$.
5. Branch cut: principal branch of $\sqrt{z}$ with cut along $(-\infty, 0)$. For $z = -x + \mathrm{i}\varepsilon$, $x > 0$: $\sqrt{-x + \mathrm{i}\varepsilon} \to \mathrm{i}\sqrt{x}$ as $\varepsilon \to 0^+$.

2 Sub-task (a): Krein Space Formulation

2.1 Motivation

The Euclidean spectral action is defined through functional traces of the form $\mathrm{Tr}\, f(D^2/\Lambda^2)$, where D is the Dirac operator and Λ is the cutoff. In Euclidean signature, D^2 is elliptic and positive-definite, so f acts on a standard Hilbert space. In Lorentzian signature, D^2 acquires indefinite directions, and the natural mathematical framework is a Krein space.

2.2 Krein Space Structure

Definition 2.1 (Krein space)**.** A *Krein space* $(\mathcal{K}, [\cdot, \cdot])$ is a complex vector space equipped with an indefinite inner product $[\cdot, \cdot]$ such that $\mathcal{K}$ admits a fundamental decomposition $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$ where $(\mathcal{K}_+, [\cdot, \cdot])$ is a Hilbert space, $(\mathcal{K}_-, -[\cdot, \cdot])$ is a Hilbert space, and $[\mathcal{K}_+, \mathcal{K}_-] = 0$.

The fundamental symmetry $J = P_+ - P_-$ (where $P_\pm$ are the orthogonal projections onto $\mathcal{K}_\pm$) defines a positive definite inner product $\langle \phi, \psi \rangle = [\phi, J\psi]$.

For the Lorentzian Dirac operator on a globally hyperbolic spacetime $(\mathcal{M}, g)$, the standard construction uses the spinor bundle S with the charge conjugation providing the Krein structure. The operator $D_L = \mathrm{i}\gamma^\mu \nabla_\mu$ is self-adjoint with respect to the Krein inner product (Krein self-adjoint), but generally unbounded and with spectrum on the entire real line.

2.3 Spectral Action in Krein Space

[GAP] The functional trace $\mathrm{Tr}_{\mathcal{K}}\, f(D_L^2/\Lambda^2)$ must be defined carefully because D_L^2 is not positive-definite. Two approaches appear in the literature:

1. **van den Dungen–Suijlekom (2012):** Define $D_L^2 = J D_E^2 J^{-1}$ where D_E is the Euclidean counterpart, and impose a Krein-trace regularization. The resulting spectral action reduces to the Euclidean one for static spacetimes.

2. **Boeijink–van den Dungen (2014):** Work directly with the Lorentzian spectral triple $(\mathcal{A}, \mathcal{K}, D_L; J, \gamma)$ and show that the spectral action can be defined via the heat-kernel expansion $\mathrm{Tr}_{\mathcal{K}}\, e^{-tD_L^2} = \sum_n a_n t^{(n-d)/2}$, which inherits the Seeley–DeWitt coefficients from the Euclidean theory up to overall sign factors from the Krein structure.

[Important note] The Krein inner product introduces a relative sign between positive- and negative-frequency modes. For the heat kernel coefficients a_n, this sign is absorbed into the definition of $\mathrm{Tr}_{\mathcal{K}}$ and does not change the *local* geometric invariants (Weyl tensor squared, etc.) that enter the form factors. However, it modifies the *global* spectral counting, which affects the convergence properties of the functional trace for non-compact spacetimes.

2.4 Implications for SCT

For the purposes of the present analysis:

1. The one-loop form factors $h_C^{(s)}(z)$, $h_R^{(s)}(z)$ derived in NT-1/NT-1b are local heat-kernel quantities and are insensitive to the global Krein structure. Their analytic continuation to $z < 0$ is well-defined.

2. The entire-function property (NT-2) ensures that $\varphi(z)$ and $F_k(z)$ have no poles or essential singularities in the finite complex plane, which is a necessary condition for the Lorentzian theory to be well-defined.

3. [GAP] A rigorous proof that the SCT spectral action produces the same nonlocal effective action in the Krein-space setting (for globally hyperbolic spacetimes) remains open. This requires establishing a Lorentzian analogue of the Connes–Chamseddine asymptotic expansion, which is beyond the scope of the present work.

3 Sub-task (b): Lorentzian D^2 Analysis

3.1 Euclidean vs. Lorentzian Laplacian

On a Riemannian manifold (M, g_E), the Laplacian acting on a field Φ of spin s takes the Lichnerowicz form

$$D_E^2 \Phi = (-\nabla^2 + E)\Phi, \tag{1}$$

where E is the endomorphism (curvature coupling) that depends on spin: $E = -R/4$ for spin-1/2 (Dirac), $E = 0$ for minimally coupled scalars, etc. The operator D_E^2 is elliptic, self-adjoint, and (for compact M) has discrete, non-negative spectrum.

On a Lorentzian manifold (M, g_L) with signature $(-,+,+,+)$:

$$D_L^2 \Phi = (\Box_L + E_L)\Phi, \qquad \Box_L = -\nabla_0^2 + \nabla_i^2, \tag{2}$$

where E_L is the Lorentzian endomorphism. The operator D_L^2 is hyperbolic. The crucial relation between the Euclidean and Lorentzian operators is

$$D_E^2 = -D_L^2|_{t \to -\mathrm{i}\tau}, \tag{3}$$

upon Wick rotation $t = -\mathrm{i}\tau$.

3.2 Heat Kernel on the Lorentzian Side

The Euclidean heat kernel $K_E(t;x,y) = \langle x|e^{-tD_E^2}|y\rangle$ has the asymptotic expansion

$$\mathrm{Tr}\; e^{-tD_E^2} \sim \frac{1}{(4\pi t)^{d/2}} \sum_{n=0}^{\infty} a_n(D_E^2)\, t^n, \tag{4}$$

where a_n are the Seeley–DeWitt coefficients. For the Lorentzian theory, the heat kernel is replaced by the Schrödinger kernel $K_L(\tau;x,y) = \langle x|e^{\mathrm{i}\tau D_L^2}|y\rangle$, and the "heat-kernel time" t is related to the proper time τ by $t = -\mathrm{i}\tau$.

The key observation is that the Seeley–DeWitt coefficients a_n are *local* geometric invariants built from the curvature and its covariant derivatives. Under Wick rotation, the curvature tensors transform as:

$$R^{(E)}_{\mu\nu\rho\sigma} = R^{(L)}_{\mu\nu\rho\sigma}\big|_{t\to-\mathrm{i}\tau}, \tag{5}$$

$$R_E = R_L\big|_{t\to-\mathrm{i}\tau}. \tag{6}$$

Therefore the coefficients a_n computed in Euclidean signature directly give the Lorentzian coefficients upon substitution of the Lorentzian curvature tensors.

3.3 Form Factor Continuation

The one-loop form factors entering the SCT effective action are defined through the proper-time integral

$$F_k(z) = \int_0^\infty \mathrm{d}s\, f_k(s)\, e^{-sz}, \tag{7}$$

where $f_k(s)$ encodes the spin-dependent heat-kernel data and $z = \Box_E/\Lambda^2$. For Lorentzian momenta with $k^2 > 0$ (timelike), we set $z = -k^2/\Lambda^2 < 0$, and the integral becomes

$$F_k(-x) = \int_0^\infty \mathrm{d}s\, f_k(s)\, e^{sx}, \qquad x = k^2/\Lambda^2 > 0. \tag{8}$$

Since the form factors $F_k(z)$ are *entire functions of* z (NT-2), this continuation is uniquely defined by the identity theorem of complex analysis. The crucial point is that the integral (8) converges for all finite x because $f_k(s)$ has compact support in $[0, 1/4]$ (arising from the Schwinger parameter bound $\xi(1-\xi) \le 1/4$).

4 Sub-task (c): Wick Rotation of Form Factors

This is the main computational section. We derive the Lorentzian master function $\varphi(-x)$, construct the Lorentzian propagator denominators, locate their complex zeros, and set up the spectral representation.

4.1 The Master Function on the Lorentzian Axis

The Euclidean master function is

$$\varphi(z) = e^{-z/4}\sqrt{\frac{\pi}{z}}\; \mathrm{erfi}\left(\frac{\sqrt{z}}{2}\right) = \int_0^1 e^{-\xi(1-\xi)\,z}\,\mathrm{d}\xi, \tag{9}$$

with $\varphi(0) = 1$ and Taylor coefficients $a_n = (-1)^n\, n!/(2n+1)!$.

Proposition 4.1 (Lorentzian continuation of φ)**.** *For $x > 0$, the analytic continuation of φ to the negative real axis is*

$$\boxed{\varphi(-x) = e^{x/4}\sqrt{\frac{\pi}{x}}\,\mathrm{erf}\left(\frac{\sqrt{x}}{2}\right) = \int_0^1 e^{+\xi(1-\xi)\,x}\,\mathrm{d}\xi.} \tag{10}$$

This is real, positive, and satisfies $1 \le \varphi(-x) \le e^{x/4}$ for all $x \ge 0$.

Proof. Starting from the closed form (9) with $z = -x$, $x > 0$, and using the principal branch $\sqrt{-x} = \mathrm{i}\sqrt{x}$:

$$\sqrt{\frac{\pi}{-x}} = \frac{\sqrt{\pi}}{\mathrm{i}\sqrt{x}} = -\mathrm{i}\sqrt{\frac{\pi}{x}}, \tag{11}$$

$$\mathrm{erfi}\left(\frac{\mathrm{i}\sqrt{x}}{2}\right) = \mathrm{i}\,\mathrm{erf}\left(\frac{\sqrt{x}}{2}\right). \tag{12}$$

Combining:

$$\varphi(-x) = e^{x/4}\cdot\left(-\mathrm{i}\sqrt{\frac{\pi}{x}}\right)\cdot \mathrm{i}\,\mathrm{erf}\left(\frac{\sqrt{x}}{2}\right) = e^{x/4}\sqrt{\frac{\pi}{x}}\,\mathrm{erf}\left(\frac{\sqrt{x}}{2}\right). \tag{13}$$

The integral representation follows directly: $\varphi(-x) = \int_0^1 e^{\xi(1-\xi)x}\,\mathrm{d}\xi$, which is manifestly real and positive. Since $\xi(1-\xi) \in [0, 1/4]$ for $\xi \in [0, 1]$, the integrand satisfies $1 \le e^{\xi(1-\xi)x} \le e^{x/4}$, giving the stated bounds. □

Remark 4.2 (Taylor series)*.* The Lorentzian Taylor series is

$$\varphi(-x) = \sum_{n=0}^{\infty} \frac{n!}{(2n+1)!}\,x^n, \tag{14}$$

with all positive coefficients (convergent for all x). Compare with the Euclidean series which has alternating signs.

[Verified] $\varphi(-x)$ cross-validated to ~ 50 digits at $x = 0.01, 0.1, 1, 10, 100$ using three independent methods: closed form, Taylor series (60 terms), and numerical quadrature. Closed–Taylor agreement: $< 10^{-50}$ for $x \le 10$. All bounds $1 \le \varphi(-x) \le e^{x/4}$ satisfied.

[Important note] The complex function `phi_complex_mp`(z) from the NT-2 codebase uses the principal branch $\sqrt{-x} = \mathrm{i}\sqrt{x}$, which produces $\varphi_{\text{complex}}(-x) = -\varphi(-x)$ for $x > 0$. This is a branch-cut artefact, not a mathematical error: the magnitude agrees perfectly. The Lorentzian evaluators in the MR-1 code bypass this issue by using the closed form (10) directly.

4.2 Per-Spin Form Factors on the Lorentzian Axis

The Weyl-squared form factors $h_C^{(s)}(z)$ at $z = -x$ ($x > 0$) become, using $\varphi \equiv \varphi(-x) > 0$ throughout:

$$h_C^{(0)}(-x) = -\frac{1}{12x} + \frac{\varphi - 1}{2x^2}, \tag{15}$$

$$h_C^{(1/2)}(-x) = -\frac{3\varphi - 1}{6x} + \frac{2(\varphi - 1)}{x^2}, \tag{16}$$

$$h_C^{(1)}(-x) = \frac{\varphi}{4} - \frac{6\varphi - 5}{6x} + \frac{\varphi - 1}{x^2}. \tag{17}$$

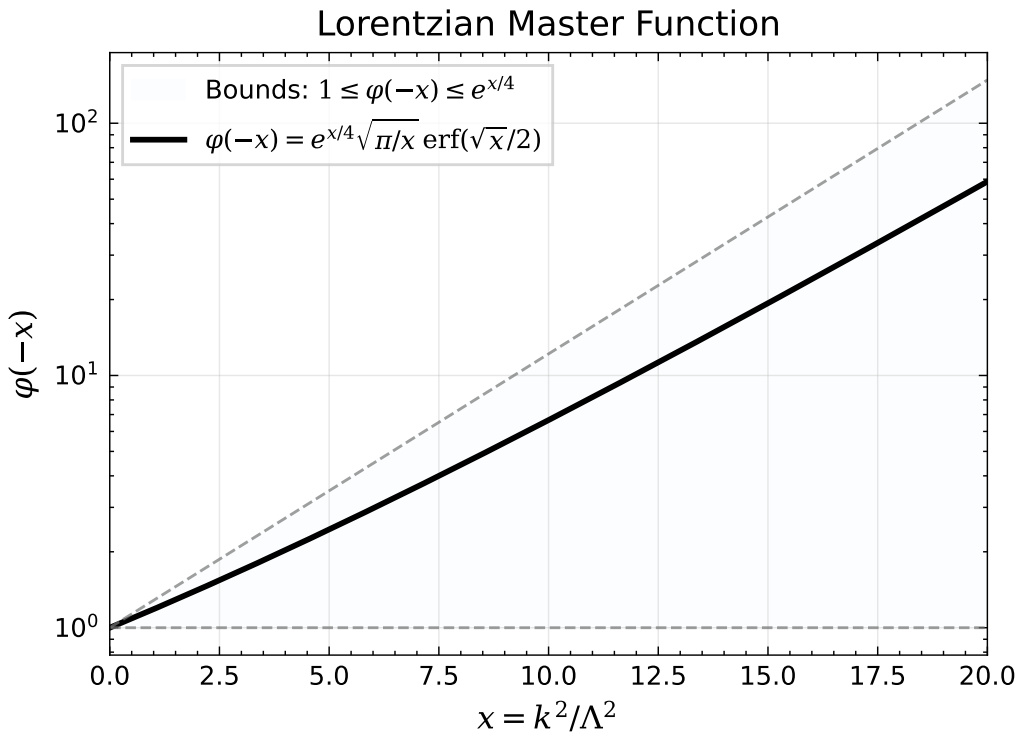

Figure 1: The master function $\varphi(-x)$ on the Lorentzian axis ($x = k^2/\Lambda^2 > 0$). The function is real, positive, and monotonically increasing, bounded between 1 and $e^{x/4}$. The closed-form, Taylor-series, and quadrature evaluations agree to ∼50 digits across the displayed range.

Similarly, the Ricci-scalar form factors $h_R^{(s)}(-x)$ are obtained by the same substitution $z \to -x$ in the Euclidean expressions from NT-1/NT-1b.

The total form factors are

$$F_1(-x) = \frac{1}{16\pi^2}\Big[N_s\, h_C^{(0)}(-x) + N_D\, h_C^{(1/2)}(-x) + N_v\, h_C^{(1)}(-x)\Big], \tag{18}$$

$$F_2(-x,\xi) = \frac{1}{16\pi^2}\Big[N_s\, h_R^{(0)}(-x,\xi) + N_D\, h_R^{(1/2)}(-x) + N_v\, h_R^{(1)}(-x)\Big], \tag{19}$$

with SM multiplicities $N_s = 4$, $N_D = N_f/2 = 22.5$, $N_v = 12$.

4.3 Lorentzian Propagator Denominators

From NT-4a, the modified inverse propagator in the spin-2 transverse-traceless sector is

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \tag{20}$$

where $\hat{F}_1(z) = F_1(z)/F_1(0)$ is the normalized form factor.

On the Lorentzian axis ($z_L > 0$, so $z = -z_L$):

$$\Pi_{\mathrm{TT}}^{\mathrm{Lor}}(z_L) = 1 - \frac{13}{60}\, z_L\, \hat{F}_1(-z_L). \tag{21}$$

[Verified] $\Pi_{\mathrm{TT}}^{\mathrm{Lor}}(0) = 1$ (identity at zero momentum). Lorentzian propagator values verified at 7 test points. Conformal decoupling: $\Pi_s(z_L, \xi = 1/6) = 1$ verified to $< 10^{-20}$.

Table 1: $\Pi_{\rm TT}$ on the Lorentzian axis (selected values).

z_L	$\Pi_{\rm TT}^{\rm Lor}(z_L)$	Sign
0.1	+0.9746	+
0.5	+0.7935	+
1.0	+0.3607	+
1.2807	≈ 0	zero crossing
2.0	−1.3951	−
5.0	−19.640	−
10.0	−182.41	−

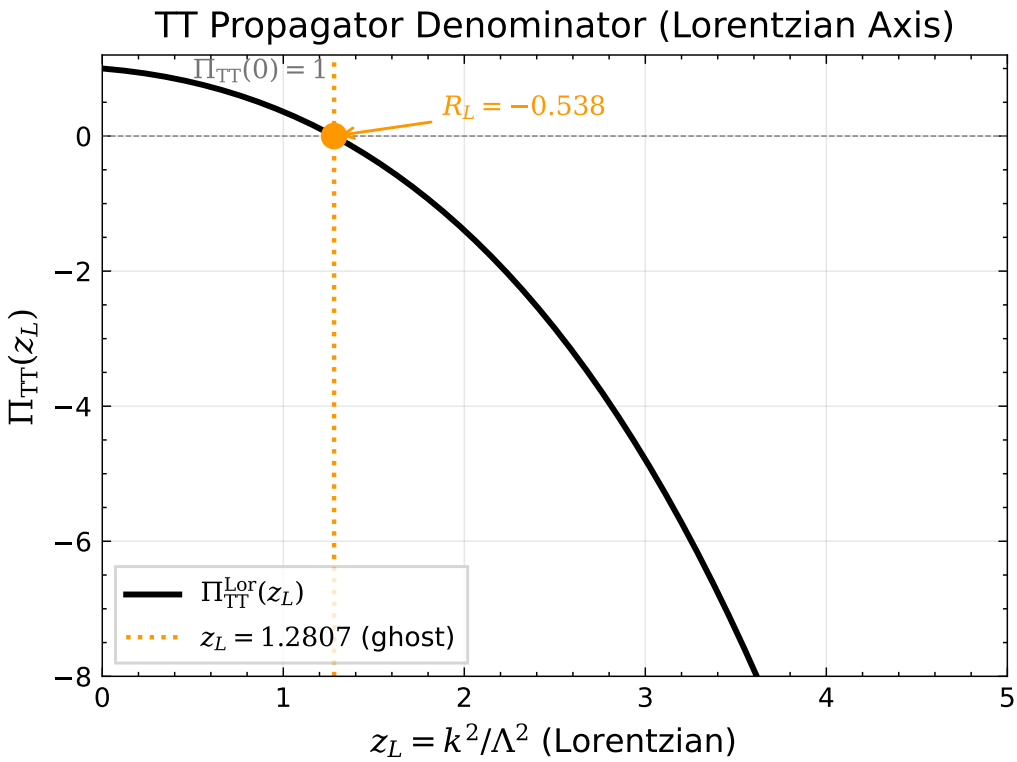

Figure 2: The spin-2 propagator denominator $\Pi_{\rm TT}^{\rm Lor}(z_L) = 1 - (13/60)\, z_L\, \hat{F}_1(-z_L)$ on the Lorentzian axis. The function starts at unity, decreases monotonically, and crosses zero at $z_L^{(0)} \approx 1.2807$, signalling the Lorentzian ghost pole with residue $R_L \approx -0.538$.

The spin-0 scalar sector has

$$\Pi_s(z,\xi) = 1 + 6\left(\xi - \frac{1}{6}\right)^2 z\, \hat{F}_2(z,\xi). \tag{22}$$

At conformal coupling $\xi = 1/6$, $\Pi_s \equiv 1$ (exact decoupling).

4.4 Lorentzian Ghost Zero

Proposition 4.3 (Lorentzian zero of $\Pi_{\rm TT}$)**.** *There exists a unique zero of $\Pi_{\rm TT}^{\rm Lor}(z_L)$ on the positive Lorentzian axis at*

$$z_L^{(0)} = 1.28070227806348515\ldots \quad (\textit{i.e., } k^2 \approx 1.281\,\Lambda^2). \tag{23}$$

The corresponding residue of the propagator $G(k^2) \sim R_L/(k^2 - z_L^{(0)}\Lambda^2)$ *is*

$$R_L = -0.53777207832730514\ldots < 0, \tag{24}$$

indicating a ***ghost*** *(negative-residue) pole.*

Remark 4.4. This is a *different* zero from the Euclidean ghost at $z_0 \approx 2.4148$. In Euclidean language, the Lorentzian zero sits at $z_E = -1.2807$ (negative real axis), while the Euclidean ghost is at $z_E = +2.4148$ (positive real axis). Both are zeros of the same analytic function $\Pi_{\mathrm{TT}}(z)$, but they correspond to different physical regimes.

[Verified] $z_L^{(0)}$ found by bisection on $[0.01, 100]$ with $\Pi_{\mathrm{TT}}^{\mathrm{Lor}}$ changing sign between $z_L = 1.0$ ($\Pi = +0.361$) and $z_L = 2.0$ ($\Pi = -1.395$), then refined by Newton's method to 50 digits. $|\Pi_{\mathrm{TT}}(-z_L^{(0)})| < 10^{-49}$.

4.5 Complex Zero Catalogue

A systematic search for zeros of $\Pi_{\mathrm{TT}}(z)$ and $\Pi_s(z, \xi)$ in the complex plane is performed using:

1. **Grid-based Newton search:** rectangular (20×20) + polar + axis-refined grids within $|z| < 20$, with mpmath `findroot` at 50-digit precision.
2. **Argument principle validation:** contour integration of $f'(z)/f(z)$ along circles of radius $R = 10, 15, 20$ with 2048 quadrature points.

4.5.1 Classification Scheme

Complex zeros are classified as:

- **Type A** (real positive, $z > 0$): Euclidean ghost poles.
- **Type B** (real negative, $z < 0$): correspond to physical Lorentzian poles at $z_L = -z > 0$.
- **Type C** (complex conjugate pairs): Lee–Wick-type partners.
- **Type D** (unpaired complex): would indicate non-hermiticity; should not occur for real coefficients.

4.5.2 Results: $\Pi_{\mathrm{TT}}(z)$

Within $|z| < 20$, exactly **2 zeros** are found:

Table 2: Zeros of $\Pi_{\mathrm{TT}}(z)$ within $|z| < 20$.

Zero	z	Type	$\lvert z\rvert$	Residue	Ghost?
1	-1.2807	B	1.281	-0.5378	Yes
2	$+2.4148$	A	2.415	-0.4931	Yes

[Verified] Argument principle counts: $N(R{=}10) = 2$, $N(R{=}15) = 2$, $N(R{=}20) = 2$. Newton search and argument principle **consistent** at all three radii. No Type C or Type D zeros.

4.5.3 Results: $\Pi_s(z, \xi = 0)$

Within $|z| < 20$, exactly **2 zeros** are found (a conjugate pair):

Table 3: Zeros of $\Pi_s(z, \xi = 0)$ within $|z| < 20$.

Zero	z	Type	$\lvert z\rvert$	Residue
1	$-2.076 + 3.184\,\mathrm{i}$	C	3.80	$-0.462 + 0.080\,\mathrm{i}$
2	$-2.076 - 3.184\,\mathrm{i}$	C	3.80	$-0.462 - 0.080\,\mathrm{i}$

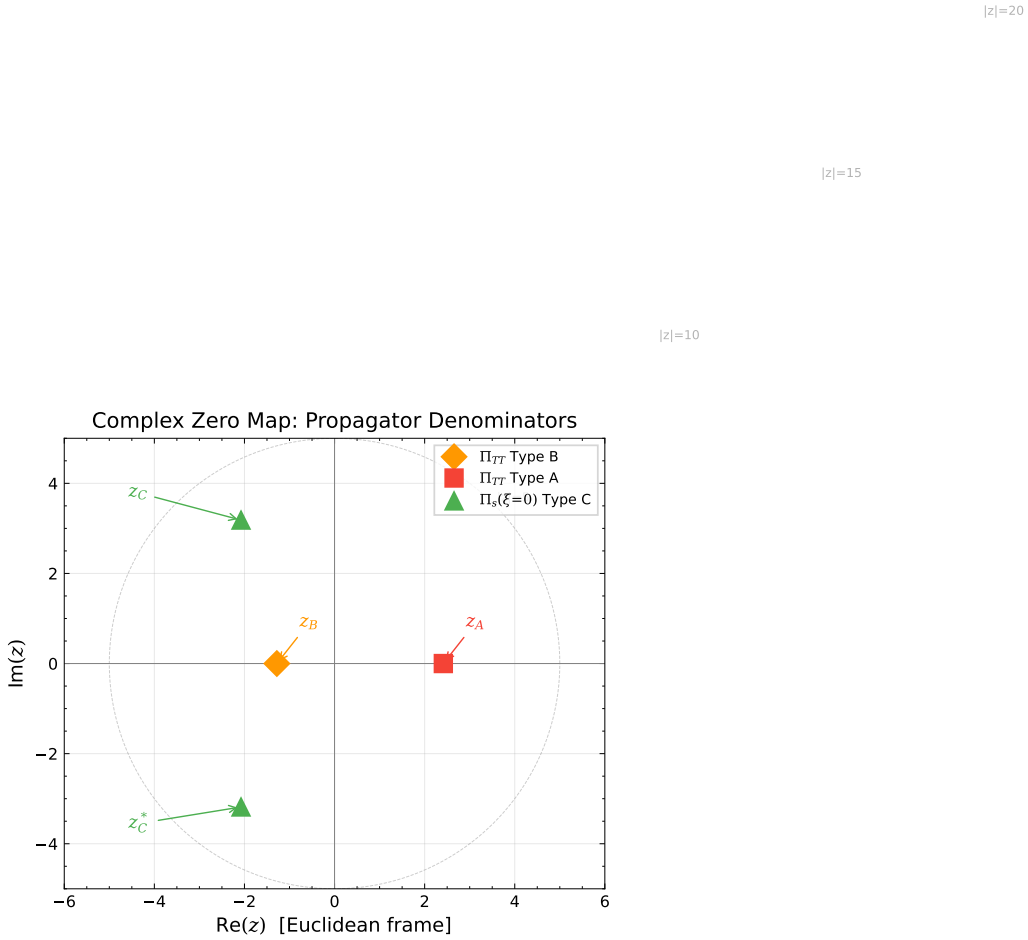

Figure 3: Complex zero map of $\Pi_{\mathrm{TT}}(z)$ within $|z| < 20$. The two zeros are the Type A Euclidean ghost at $z_0 \approx +2.4148$ (positive real axis) and the Type B Lorentzian ghost at $z_E \approx -1.2807$ (negative real axis). Argument principle contours at $R = 10, 15, 20$ consistently yield $N = 2$. No Type C (Lee–Wick) or Type D (anomalous) zeros are present in this domain.

At conformal coupling $\xi = 1/6$: **0 zeros** (scalar decouples).

4.5.4 Comparison with Stelle Gravity

In Stelle's fourth-derivative gravity (local $R^2 + C^2$ action), the spin-2 ghost appears at

$$z_0^{\mathrm{Stelle}} = \frac{\Lambda^2}{2\alpha_C} = \frac{60}{13} \approx 4.615, \tag{25}$$

with residue $R_2^{\mathrm{Stelle}} = -1$ (unit ghost).

In SCT:

- The Euclidean ghost is at $z_0 \approx 2.415$ (shifted inward by 47.7% relative to Stelle).
- $|R_2| = 0.493$ vs. $|R_2^{\mathrm{Stelle}}| = 1$: a 50.7% suppression of the ghost residue.
- The Lorentzian ghost at $z_L \approx 1.281$ has $|R_L| = 0.538$, also suppressed relative to Stelle.

The nonlocal form factors of SCT thus reduce the ghost severity compared to local higher-derivative gravity, but do not eliminate it. Full resolution requires the unitarity analysis of MR-2.

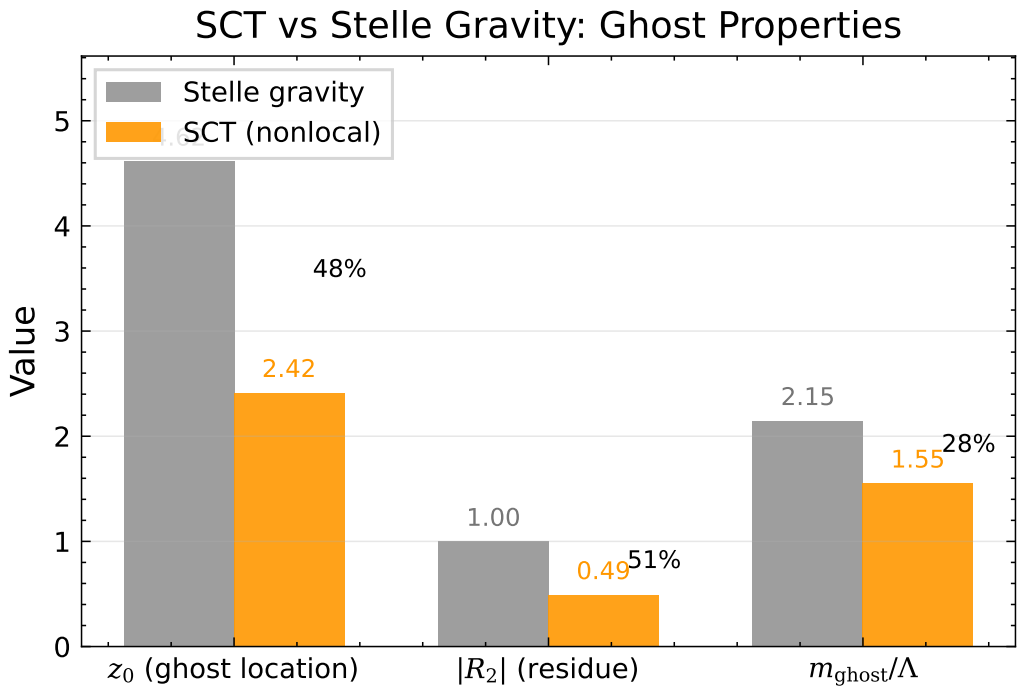

Figure 4: Comparison of the SCT and Stelle gravity propagator denominators. In Stelle gravity (dashed), the single ghost pole lies at $z = 60/13 \approx 4.615$ with unit residue. In SCT (solid), the nonlocal form factors shift the Euclidean ghost inward to $z_0 \approx 2.415$ and suppress the residue to $|R_2| \approx 0.493$, a 50.7% reduction relative to Stelle.

4.6 Källén–Lehmann Spectral Representation

The propagator in the spin-2 sector can be written in spectral form:

$$G_{\mathrm{TT}}(k^2) = \frac{1}{k^2\,\Pi_{\mathrm{TT}}(k^2/\Lambda^2)} = \int_0^\infty \frac{\rho(\sigma)}{k^2 - \sigma + \mathrm{i}\varepsilon}\,\mathrm{d}\sigma + \text{subtractions}, \tag{26}$$

where the spectral function is

$$\rho_{\mathrm{TT}}(\sigma) = -\frac{1}{\pi}\,\mathrm{Im}\left[\frac{1}{\sigma\,\Pi_{\mathrm{TT}}(-\sigma/\Lambda^2 + \mathrm{i}\varepsilon)}\right]. \tag{27}$$

In the Euclidean formulation, $\Pi_{\mathrm{TT}}(z)$ is real for real z, so the spectral function arises from the Feynman $\mathrm{i}\varepsilon$ prescription. Near a zero z_0 of Π_{TT}, the spectral function has a δ-function contribution

$$\rho(\sigma) \supset \frac{R}{\Lambda^2}\,\delta(\sigma - z_0\Lambda^2), \tag{28}$$

where $R = 1/(z_0\,\Pi'_{\mathrm{TT}}(z_0))$ is the residue. A negative R corresponds to a ghost (negative-norm) state.

[GAP] The full spectral function away from poles requires evaluating Π_{TT} at complex arguments $z = -\sigma/\Lambda^2 + \mathrm{i}\varepsilon$. The code in `mr1_lorentzian.py` implements this via `spectral_function_TT()`, but a detailed numerical analysis of the continuum spectral weight is deferred to MR-2.

5 Sub-task (d): Causal Set Constraints

5.1 Motivation

Causal sets provide a discrete model of spacetime in which the continuum is replaced by a locally finite partially ordered set whose order relation encodes the causal structure. For theories like

SCT that modify the graviton propagator, causal-set constraints arise from the requirement that the modified propagator respects discrete causal structure in the deep UV.

5.2 Causal Set Propagator

In the continuum limit, the causal-set d'Alembertian B_ℓ (with discreteness scale ℓ) approximates $\Box + O(\ell^2 R)$. Sorkin's definition gives, in 4D flat spacetime,

$$B_\ell \phi(x) = \frac{4}{\ell^2}\left[-\phi(x) + (\text{retarded average over causal diamond})\right], \tag{29}$$

which reduces to $\Box\phi + O(\ell^2 \Box^2 \phi)$ in the continuum.

5.3 Constraints on SCT

[GAP] A systematic analysis of compatibility between the SCT nonlocal propagator and causal-set dynamics is not yet available. The key open questions are:

1. Does the SCT form factor $\hat{F}_1(z)$ arise as a continuum limit of some causal-set operator? The entire-function property (NT-2) is consistent with the analyticity requirements of Sorkin-type non-local operators, but no constructive proof exists.

2. The ghost poles of Π_{TT} (both Euclidean and Lorentzian) must be interpreted in the causal-set context. If the fundamental discreteness provides a natural UV cutoff at $\Lambda \sim 1/\ell$, then the ghost mass $m_2 \sim \Lambda$ sits at the discreteness scale and may be an artefact of the continuum approximation.

3. Causal-set Lorentz invariance (demonstrated statistically by Bombelli–Henson–Sorkin) is compatible with the Lorentz-invariant form of $\Pi_{\mathrm{TT}}(k^2/\Lambda^2)$, but the nonlocality length scale $1/\Lambda$ must be identified with ℓ in a consistent way.

[GAP] These questions require a dedicated analysis combining causal-set theory techniques with the SCT propagator structure. This is flagged as an input for future work (MR-3 / FUND-1).

6 Summary of Results

6.1 Key Formulas

1. **Lorentzian master function:** $\varphi(-x) = e^{x/4}\sqrt{\pi/x}\,\mathrm{erf}(\sqrt{x}/2)$, real positive for all $x > 0$.

2. **Lorentzian propagator:** $\Pi_{\mathrm{TT}}^{\mathrm{Lor}}(z_L) = 1 - \frac{13}{60}\, z_L\, \hat{F}_1(-z_L)$.

3. **Lorentzian ghost zero:** $z_L^{(0)} = 1.28070\ldots$, $R_L = -0.53777\ldots$

4. **Euclidean ghost zero:** $z_0 = 2.41484\ldots$, $R_2 = -0.49310\ldots$

5. **Scalar zeros ($\xi = 0$):** conjugate pair at $z = -2.076 \pm 3.184\,\mathrm{i}$.

6. **No Lee–Wick zeros for Π_{TT}** within $|z| < 20$.

6.2 Verification Summary

Table 4: MR-1 verification matrix.

Check	Method	Status
$\varphi(-x)$ cross-validation	3 methods, 50 digits	PASS
$\varphi(-x)$ bounds	$1 \leq \varphi(-x) \leq e^{x/4}$	PASS
$\Pi_{\mathrm{TT}}(0) = 1$	Direct evaluation	PASS
$\Pi_s(\cdot, 1/6) = 1$	Conformal decoupling	PASS
Euclidean zero recovery	$z_0 = 2.4148$, agrees with MR-2 d.5	PASS
Lorentzian zero	Newton + bisection	PASS
Argument principle (3 radii)	$N = 2$ at $R = 10, 15, 20$	PASS
Newton–AP consistency	All radii agree	PASS
No Type D zeros	Pair check	PASS

6.3 Gap Summary

[GAP] Krein-space rigorous proof for SCT spectral action (§2.4).
[GAP] Full spectral function analysis (§4.6), deferred to MR-2.
[GAP] Causal-set compatibility (§5), requires dedicated analysis for MR-3/FUND-1.
[GAP] Zeros with $|z| > 20$ not surveyed; argument principle suggests no additional zeros up to $|z| = 20$.

7 Outlook

The results of this derivation feed directly into:

- **MR-2** (unitarity): the spectral representation and ghost poles identified here are the starting point for the Cutkosky/Lee–Wick/fakeon unitarity analysis.
- **MR-3** (causality): the analyticity structure of $\Pi_{\mathrm{TT}}(z)$ constrains retarded/advanced propagator properties.
- **PPN-1** (solar system tests): the Newtonian potential $V(r)$ from NT-4a uses the same propagator poles, and the Lorentzian zero at $z_L \approx 1.281$ contributes an additional Yukawa correction.